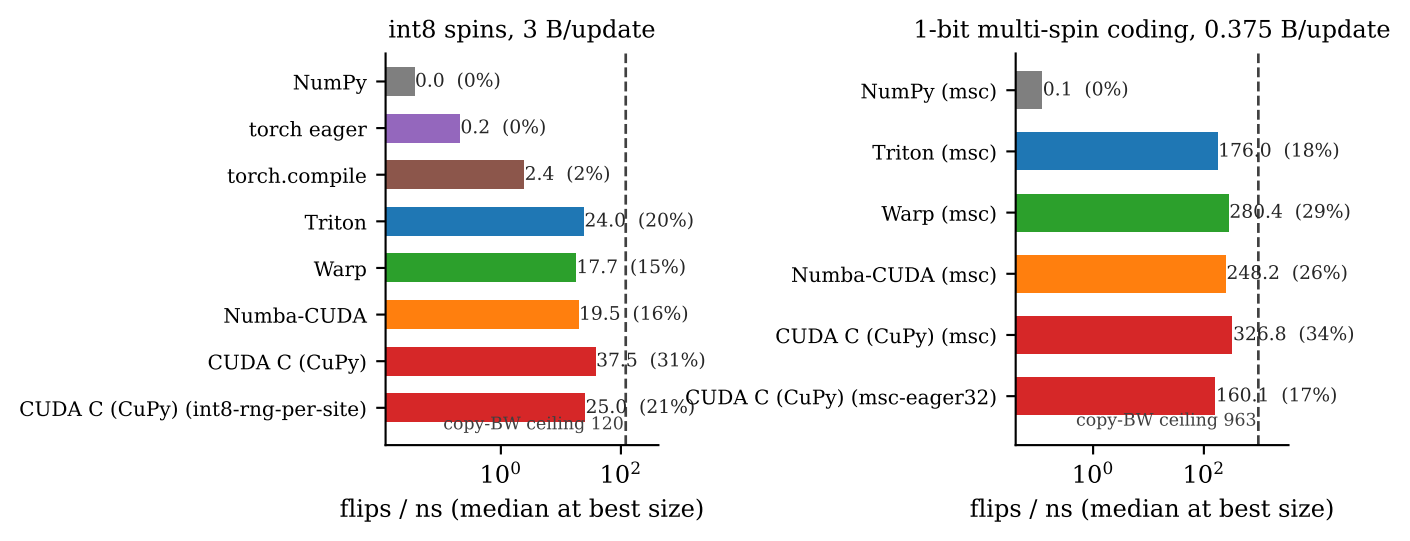
int8 spins, 3 B/update
NumPy
torch eager
torch.compile
Triton
Warp
Numba-CUDA
CUDA C (CuPy)
CUDA C (CuPy) (int8-rng-per-site)
0.0 (0%)
0.2 (0%)
2.4 (2%)
24.0 (20%)
17.7 (15%)
19.5 (16%)
37.5 (31%)
25.0 (21%)
copy-BW ceiling 120
10^0
10^2
flips / ns (median at best size)
1-bit multi-spin coding, 0.375 B/update
NumPy (msc)
Triton (msc)
Warp (msc)
Numba-CUDA (msc)
CUDA C (CuPy) (msc)
CUDA C (CuPy) (msc-eager32)
0.1 (0%)
176.0 (18%)
280.4 (29%)
248.2 (26%)
326.8 (34%)
160.1 (17%)
copy-BW ceiling 963
10^0
10^2
flips / ns (median at best size)